DLP ASSIGNMENT 2

Colorization as regression:

1) There are 6 convolutional layers.
2) The kernel size is 3x3 for all layers with 32 no. of filters
   a)

Layer name	Input channel	Output channel	Kernel size	No. of filtres
downconv1	1	32	3x3	32
downconv2	32	64	3x3	64
rfconv	64	64	3x3	64
upconv1	64	32	3x3	32
upconv2	32	3	3x3	3
finalconv	3	3	3x3	3

3) The CNN model was trained for 25 epochs with adam as its optimiser.

   Training Variations:

10 epochs	Train loss = 0.0114 , Val loss = 0.0114
25 epochs	Train loss = 0.0088 ,Val loss = 0.0085
50 epochs	Train loss = 0.0075 ,Val loss = 0.0075
75 epochs	Train loss = 0.0062 ,Val loss = 0.0064

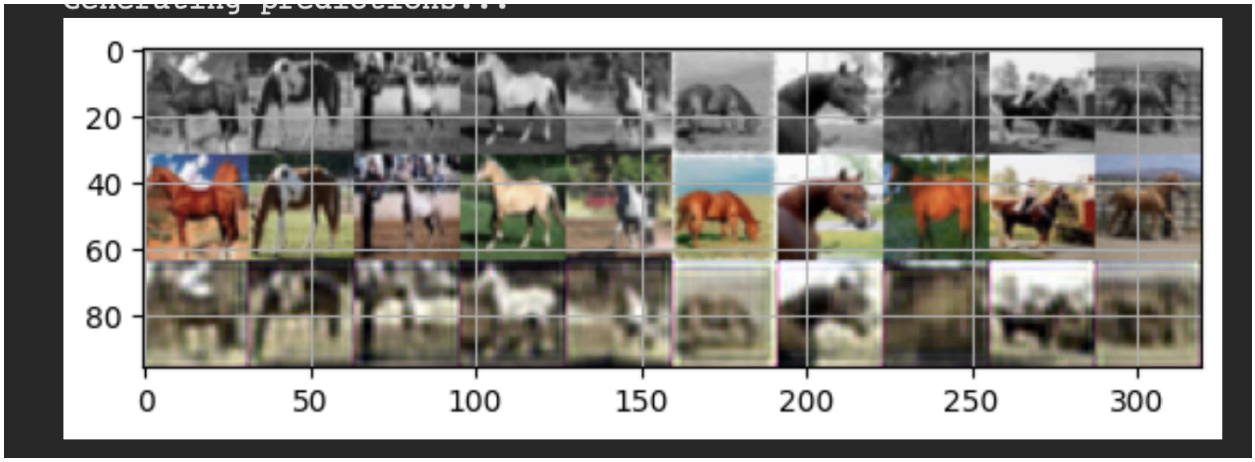
The output images became clearer as number of epochs increased form 10-50 after that the model started overfitting as val loss at 75epochs > training loss at 75epochs

The above table shows that as the number of epochs increases the validation loss significantly increases as compared to the training loss showing that the model has started overfitting as the model has started to learn the dataset because of training for too long or due to too many epochs. The optimal stopping point being 25-50 epochs.

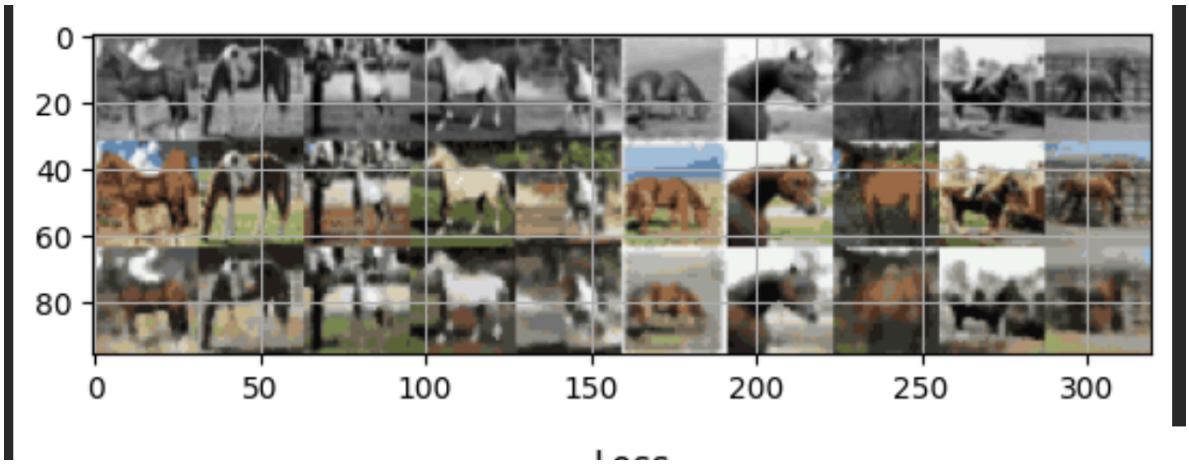
Colorisation as Classification:

Regression vs. Classification:

Reg:



Class:



Classification is giving 50% validation accuracy , howeever since the two models : REGRESSION(MSE) , CLASSIFICATION(CROSSENTROPY) measure things differently their magnitudes cannot be compared.

The classification model creates more vibrant, realistic images than the regression model because it chooses specific colors rather than averaging them into a dull gray.

While the regression model produces smoother color transitions, it lacks saturation. In contrast, the classification model delivers bold colors but suffers from color blocking , which is a banding effect caused by its limited 24-color palette.

The 50% accuracy score doesn't tell the whole story, as "wrong" guesses are often visually close to the correct color. The best example of this difference is a white horse: the regression model incorrectly predicts brown, while the classification model correctly identifies it as light. This proves that treating colorization as a classification task is superior to using regression.

SKIP CONNECTIONS:

1)

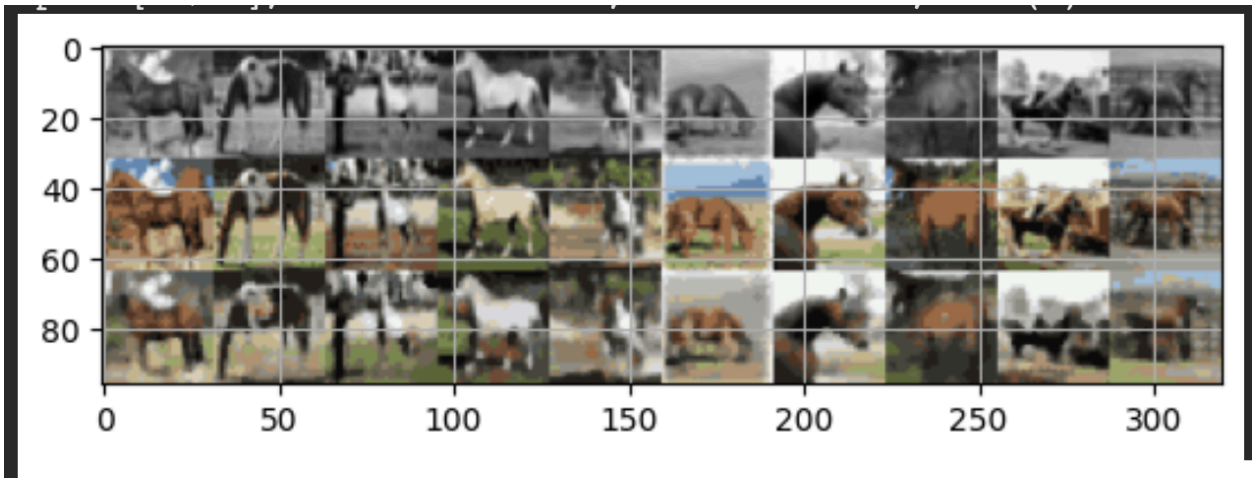
CNN: train loss = 1.4815 , validation loss = 1.4779 , val accuracy = 44.3%

UNet: train loss = 1.3066 , val loss = 1.3193 , val acc = 49.2%

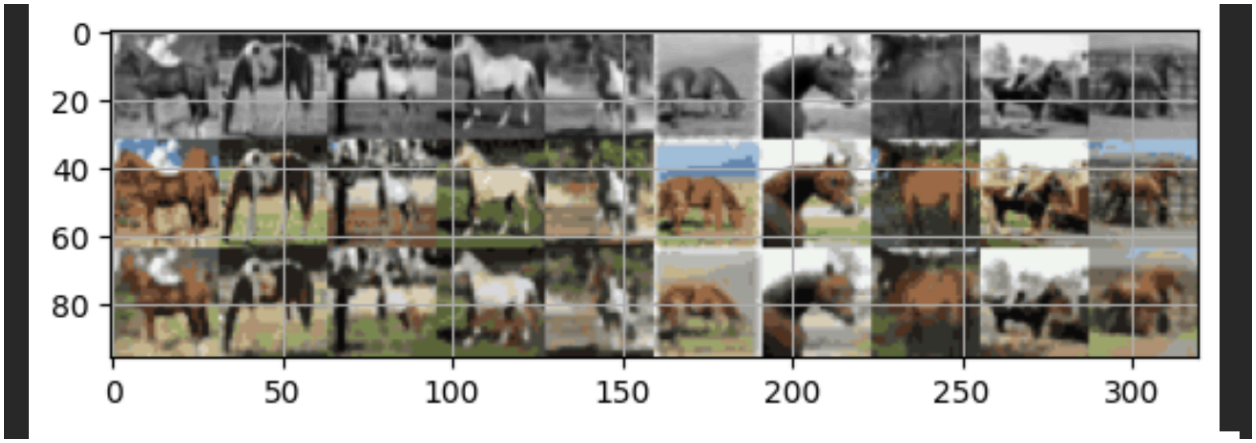
Yes , the skip connections is achieving lower training and validation accuracy as compared to CNN and has significantly higher validation accuracy across all epochs.

2)

CNN



UNet:



UNet shows better outputs with margins being modest at 32x32. The improvements are consistent , moreover , the quantitative gap of the losses and accuracies are a better way of inspecting rather than the visual inspection at this resolution.

3)

- As gradients become extremely small during backpropagation , it makes learning difficult. However, skip connections makes it easier for gradient to flow from later layers to earlier ones preserving the signal and enabling effective training.
- Skip connections let the network bypass some layers and combine low-level features with high-level representations. This encourages the model to learn residual mappings (i.e., only the difference from the input), which often makes optimization easier and improves accuracy.

Visualizing intermediate activations:
1:

Conv1: Fine-grained local detectors
All 32 feature maps at 16×16 resolution are densely activated. Each channel responds to specific low-level patterns, such as curved backs, vertical legs, or flat background areas. The overall horse shape is easily recognizable at this stage. This layer retains the most spatial detail.

Conv2: More abstract features
The resolution reduces to 8×8. About half of the feature maps are mostly inactive, while the active ones capture broader patterns, representing entire body regions rather than individual edges. The horse silhouette is still visible, but now more like a rough mask than detailed structure.

Conv3: Information bottleneck
At this stage, no clear horse shape remains. The network has compressed the image into a small set of abstract cues that signal general concepts like foreground vs background or textured vs smooth regions. All subsequent color predictions must pass through this narrow channel.

Later layers: Semantic recovery
After the bottleneck, feature maps begin representing broader semantic regions. Although boundaries remain diffuse, the network starts reconstructing meaningful areas from the compressed representation.

Conv5: Color probability maps
At 32×32 resolution, each of the 24 channels represents a probability map for a specific color cluster. The final color prediction is obtained by taking the argmax across these channels.

2:

In a standard CNN, the early encoder layers (conv1 and conv2) still detect edges and shapes, but the bottleneck layer (conv3) becomes extremely sparse. Most feature maps display only a few bright spots, losing the horse's outline entirely. The decoder must reconstruct color from this limited information, leading to blurry and bleeding boundaries.

In a UNet, the early encoder layers look similar, but with a crucial difference: because the decoder receives these encoder features directly through skip connections, the network doesn't need to force all spatial detail through the bottleneck. As a result, the bottleneck maps (conv3) are cleaner, focusing on global semantics rather than precise locations. The decoder layers show dramatic improvement: by conv4, region boundaries are much sharper, and in conv5—the final 24 color probability maps—the colors are nearly binary, with one clearly inside the horse and another outside, creating crisp separation.

The key takeaway is that skip connections allow the network to separate “what” from “where”: the bottleneck manages global color decisions, while the encoder features provide exact edge information to the decoder, resulting in sharper activations and cleaner color boundaries.

Theoretical questions:

1. Pixel-wise loss functions (such as MSE or cross-entropy) evaluate images by comparing individual pixels, but they ignore whether the image appears realistic overall. As a result, an image can have a low pixel error yet still look blurry and unappealing to the human eye.A more effective alternative is perceptual loss. Instead of directly comparing pixel values, both the predicted and ground truth images are passed through a pre-trained network like VGG, and their high-level feature representations are compared. These features capture important aspects such as structure, texture, and semantic meaning—elements that align more closely with human perception. Optimizing perceptual loss typically produces sharper and more visually appealing results.Additionally, evaluation metrics like SSIM (Structural Similarity Index) and LPIPS (Learned Perceptual Image Patch Similarity), or even user studies, can be used to better reflect perceived image quality.

2. The model was trained on 32×32 patches, but since it is fully convolutional and contains no fully connected layers, it can handle images of any size as long as spatial dimensions are managed properly.One approach is to move a 32×32 window across a larger image, colorize each patch, and then combine the outputs, averaging overlapping regions to prevent visible seams.Alternatively, because all operations (convolution, pooling, and upsampling) are spatially flexible, the entire image can be passed through the network at once, producing an output of the same size. However, it is important to ensure compatibility between downsampling and upsampling. If the input dimensions are not multiples of the total downsampling factor (4 in this case), padding or resizing may be required. In practice, padding the image to a multiple of 4, processing it, and then cropping it back is a simple solution.